

A new, highly potent 1,8-naphthalimide-based fluorescence turn-off chemosensor capable of detecting Cu(II) ions in real-world water samples

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The synthesis of a new 1,8-naphthalimide-based fluorescence turn-off chemosensor for the effective detection of Cu(II) ions is reported. The chemosensor MAST was characterised by mp, Fourier Transform-infrared (FTIR), ultraviolet-visible (UV-vis), fluorescence, $^1\text{H-NMR}$ and $^{13}\text{C-NMR}$ spectroscopy, and mass spectrometry. After treatment with Cu(II) ions in coexistence with various competitive metal ions in a dimethylsulphoxide (DMSO)/4-(2-hydroxyethyl)-1-piperazineethanesulphonic acid (HEPES) buffer solution (v/v, 1:1; pH 7.4), the chemosensor demonstrated high selectivity towards Cu(II) ions and a fluorescence quenching of 83.67% was observed. In addition, a good linear response for the chemosensor detecting Cu(II) ions was obtained over the concentration range 0.2–1 μM . The combined results of UV-vis and fluorescence titrations with the Job plot yielded a 1:1 stoichiometric interaction of the chemosensor with Cu(II) ions. The association constant and detection limit were calculated as $1.37 \times 10^6 \text{ M}^{-1}$ and $1.69 \times 10^{-8} \text{ M}$, respectively. Based on results, the quenching response of MAST towards Cu(II) ions could be ascribed to the reverse-photoinduced electron transfer (PET) mechanism. Finally, the proposed chemosensor was applied to quantify Cu(II) ions in real-world water samples and excellent recovery rates of 98–105.8% were obtained.

Introduction

In recent decades, efforts have been made to develop chemosensors to monitor metal ions of biological and environmental importance [1–30]. Despite the crucial roles metal ions play in environmental and biological systems, their ecotoxicological impingements have constituted a considerable challenge [31,32]. Attention has been focused on developing chemosensors specifically for Cu(II) ions because of their importance as catalyst co-factors for certain key proteins involved in the growth and development of the human body [33–35]. Abnormal homeostatic thresholds of Cu(II) ions are linked to neurodegenerative conditions and diseases such as Wilson's disease, dyslexia, hypoglycemia, Alzheimer's disease, Menkes disease, Parkinson's disease, amyotrophic lateral sclerosis, infant liver damage and gastrointestinal disease [36–42]. Following studies of toxicity data, the World Health Organisation (WHO) and US Environmental Protection Agency (EPA) set the acceptable levels of Cu(II) ions in drinking water at 31.5 and 20 μM , respectively [35,43]. Therefore, the selective monitoring of Cu(II) ions in both biological and environmental systems is especially relevant.

Earlier methods of metal ion detection include atomic absorption spectrometry, inductively coupled plasma-atomic emission spectrometry, and inductively coupled plasma-mass spectrometry. Despite their various merits, the high cost of equipment, time-consuming nature and demanding procedures involved placed restrictions on the application of these methods [44,45]. By comparison, the fluorescence technique features high selectivity and sensitivity, simplicity of operation, a comparatively low instrumentation cost and quick analytical detection, thereby

making it a much better sensing technique [46,47]. 1,8-naphthalimide derivatives have been widely exploited as fluorophores due to their excellent photophysical fluorescent properties of high photostability and quantum yield, strong absorption band in the visible region and large Stokes shifts [48]. Consequently, 1,8-naphthalimide derivatives enjoy a wide range of applications as colorants in the polymer industry [49,50], laser dyes [51,52], chemosensors [53,54] and fluorescence probes for biomedical purposes, including as fluorescence cellmakers [55] and anti-cancer agents [56].

Although there have been significant advances in the design and synthesis of Cu(II)-selective fluorescence turn-off chemosensors [57–66], there is still scope for further progress. In this study, the synthesis of a new fluorescence turn-off chemosensor for the selective detection of Cu(II) ions, which employs the fluorophore-spacer-receptor convention based on the PET mechanism [67], is reported. Interestingly, the chemosensor displayed good sensing properties towards Cu(II) ions among other metal ions during testing, and it can also be used for monitoring Cu(II) ions in real-world environmental systems.

Experimental

Reagents and materials

All the reagents and solvents used for organic syntheses were purchased commercially and, unless otherwise stipulated, were of analytical reagent grade and used without further purification. Double-distilled water was used in all syntheses and deionised water was used for all spectroscopic measurements. Sixteen nitrate and chloride metal salts were used as the sources of the metal ions, namely,



sodium nitrate, potassium nitrate, calcium chloride, magnesium chloride, aluminium nitrate, lead nitrate, iron nitrate, nickel nitrate, copper nitrate, mercury nitrate, silver nitrate, cobalt nitrate, manganese nitrate and cadmium nitrate. HEPES (pH 7.4) was used as the buffer solution. All experiments were conducted at 20–25 °C.

Instrumentation

C, H and N content was determined using a Carlo Erba 1106 elemental analyser. The progress of all reactions were periodically monitored by thin layer chromatography (TLC) performed on silica gel (Fluka F60 254, 20 × 20, 0.2 mm) (Merck, Germany). Mp was determined by XD-4 digital micro apparatus (Beijing Tech Instrument Co., Ltd, China) and remained uncorrected. The FTIR spectra were recorded in the 4000–400 cm⁻¹ region with a Varian Scimitar 1000 spectrometer (Varian, USA) using potassium bromide pellets. Electrospray ionisation mass spectra (ESI-MS) were obtained on a Bruker micrOTOF-Q system (Bruker, Belgium). UV-vis spectra were recorded using a LabTech BlueStar Plus UV-vis spectrophotometer (Beijing LabTech Instruments Co., Ltd, China). Fluorescence spectra were measured using a Lengguang Tech. F97 Pro spectrofluorometer (Shanghai Lengguang Technology Co., Ltd, China). All the absorption and fluorescence spectra were acquired at room temperature in a 1 cm × 1 cm quartz cuvette (3 ml) using a bandwidth of 10 nm and a scan rate of 3000 nm min⁻¹. Both ¹H-NMR and ¹³C-NMR spectra were recorded on Varian Mercury plus 400 spectrometers (Varian, USA) with tetramethylsilane (TMS) as the internal standard and DMSO-d₆ as the solvent. Data, graphs and chemical structures were analysed, plotted and drawn using Microsoft Excel, Origin 8.5, Mnova NMR, ChemDraw Ultra 12.0 and Omnic software. The pH values of the metal ion solutions used were adjusted with a Beckman Coulter F360 pH meter (USA) using 0.1 M hydrogen chloride and/or 0.1 M sodium hydroxide.

Procedures for spectroscopic measurements

Metal ions of biological and environmental importance were used to monitor the changes in the fluorescence emission and absorption spectra of the synthesised chemosensor, MAST. Sixteen metal ions, Na(I), K(I), Ca (II), Mg(II), Al(III), Pb(II), Fe(III), Ni(II), Zn(II), Hg(I), Ag(I), Co(II), Mn(II), Cd(II) and Cu(II), were utilised for the spectroscopic investigation. Stock solution (1 mM) of MAST was prepared in DMSO/HEPES buffer solution (v/v, 1:1; pH 7.4) and then the sample solution of MAST was obtained by diluting the stock solution to the required molar concentration. Similarly, sample solutions of each of the metal ions were obtained by dissolving appropriate amounts of each of the metal salts in deionised water until the required concentration was reached. The absorption and fluorescence spectra of the chemosensor–metal ion solutions were recorded after completely mixing for 1 min. The fluorescence measurements were carried out at the maximum excitation wavelength of 435 nm and the maximum emission wavelength of 750 nm; the absorbance measurements took place over the wavelength range of 180–550 nm.

Synthesis

The proposed fluorescent chemosensor MAST was devised through a stepwise synthetic route (Scheme 1). The starting

point of the synthesis was 4-bromo-1,8-naphthalic anhydride (compound **1**). Intermediate compounds **2** and **3** were synthesised by a slight modification of the procedures reported by Misra *et al.* [1].

Synthesis of intermediate compound 2

A suspension of compound **1**, 4-bromo-1,8-naphthalic anhydride (8.30 g, 0.03 mol) dissolved in 80 ml of ethanol was added to benzylamine (5 ml, 0.05 mol) mixed with 20 ml of ethanol. This was then refluxed with stirring for 4 h at a temperature of 70 °C. The progress of the reaction was monitored by TLC analysis using ethylacetate/hexane (v/v, 2:8) as the eluent. Once the reaction was complete, 250 ml of cold water was added to the mixture to facilitate the precipitation of the desired product, which was then isolated by vacuum filtration, washed with 50 ml of 10% aqueous sodium carbonate and 50 ml of distilled water. Finally, the product was dried at 60 °C in a vacuum and an off-white solid was obtained; yield: 90%; mp: 140–142 °C.

Synthesis of intermediate compound 3

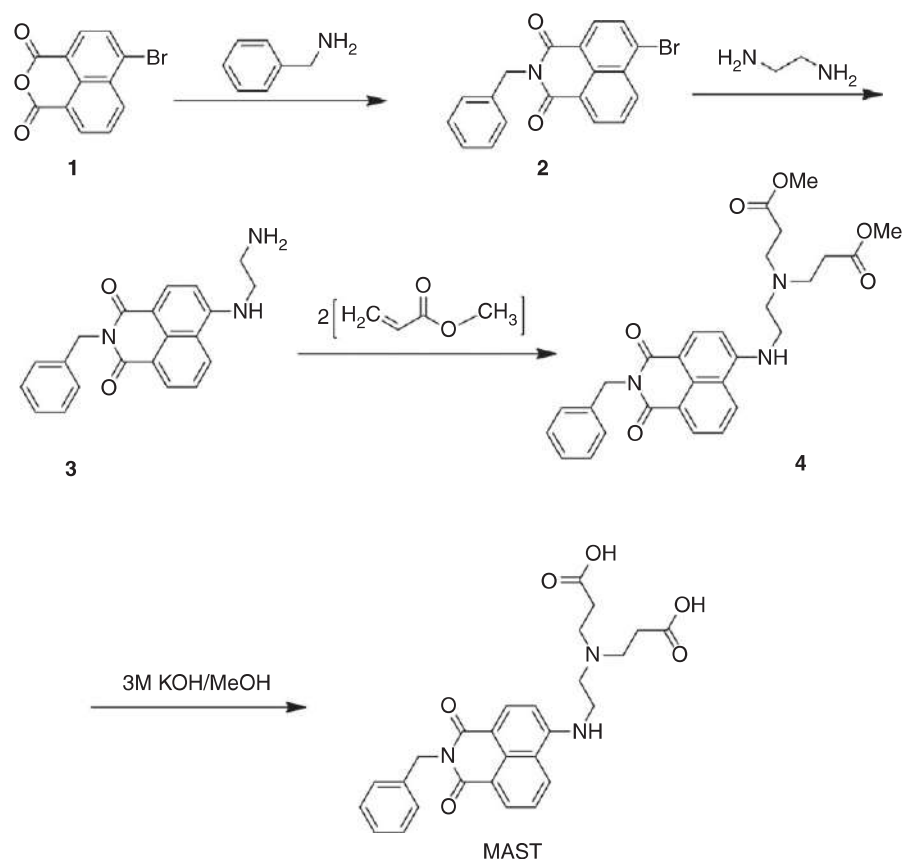
A solution of ethylenediamine (11 g, 0.183 mol) in a two-necked 500 ml round-bottomed flask was heated to 65 °C on a reflux apparatus. Next, intermediate compound **2** (1 g, 0.0027 mol) was added to the flask. The resultant mixture was heated to reflux for 4 h and stirred at 65 °C. TLC monitoring of the reaction took place hourly using 2 ml of methanol as the eluent. Once the reaction was complete, the mixture was allowed to cool. Then 100 ml of distilled water was poured into the solution, and the precipitate formed was collected by suction filtration, washed with 20 ml of distilled water and dried at 60 °C in a vacuum. The purification of the crude product was conducted with column chromatography on a silica gel stationary phase with methanol as the eluent (*R_f* = 0.25); yield of the pure product: 76%; mp: 169–170 °C.

Synthesis of intermediate compound 4

To a solution of intermediate compound **3** (1 g, 0.0029 mol) in 60 ml of methanol, a solution of methyl acrylate (2.5 g, 0.029 mol) in 10 ml of methanol was added dropwise for 30 min. The reaction mixture was stirred for five days. Once the reaction was complete, the product formed was collected. Meanwhile, excess methyl acrylate was removed by suction filtration, and the ester-functionalised derivative **4** was obtained as a yellow solid and dried at 60 °C in a vacuum; yield: 81%; mp: 178–179 °C.

Synthesis of the proposed chemosensor, MAST

The desired compound MAST was obtained in a quantitative yield of 90% by the hydrolysis of intermediate compound **3** with potassium hydroxide in a methanol solvent under reflux conditions for 2–3 h at a temperature of 75 °C; this was followed by acidification via the dropwise addition of 1 M hydrogen chloride until the pH reached 2–3, at which point a red precipitate of the desired product formed. This was then collected by vacuum filtration and washed thoroughly with 10 ml of water. The crude product was finally recrystallised in ethanol and the pure product afforded an 77% yield; mp: 167–168 °C; MS (ES⁺) *m/z* = 490.1853 (M + H)⁺; FTIR (*v*_{max}, KBr, cm⁻¹): 757 (*v*_{Ar}); 3381 (*v*_{NH}); 2603 and 2592 (*v*_N); 2871 and 2960 (*v*_{CH});



Scheme 1 Synthetic path to chemosensor MAST and reference compounds **2** and **3**

1684 ($\nu^{\text{as}}\text{N}-\text{C}=\text{O}$); 1646 ($\nu^{\text{s}}\text{N}-\text{C}=\text{O}$); 1717 ($\nu\text{C}=\text{O}$ of COOH); 3663 and 3669 (νOH of COOH); 1550 and 1582 ($\nu\text{C}=\text{C}$, aromatic); 1300 and 1350 ($\nu\text{C}-\text{O}$ of COOH). UV-vis (in DMSO, nm): 259, 445. Elemental analysis for MAST $\text{C}_{27}\text{H}_{27}\text{N}_3\text{O}_6$ (MW 489.5), analytically calculated (%): C, 66.25; H, 5.56; N, 8.58; found (%): C, 66.35; H, 5.54; N, 8.71. ^1H -NMR (300 MHz, $\text{DMSO}-d_6$): δ = 8.57 (d, 1H, J = 7.3 Hz, naphthalimide H-5); 8.45 (d, 1H, J = 8.6 Hz, naphthalimide H-2); 8.38 (dd, 1H, J = 8.3 Hz, J = 1.0 Hz, naphthalimide H-7); 7.60 (dd, 1H, J = 8.3 Hz, J = 7.3 Hz, naphthalimide H-6); 6.67 (d, 1H = 8.6 Hz, naphthalimide H-3); 6.28 (m, 1H, NH); 4.16 (t, 2H, J = 7.6 Hz, $(\text{CO})_2\text{NCH}_2$); 11.0 (s, 2H, 2 $\times$ OH); 2.39 (t, 4H, J = 6.4 Hz, 2 $\times$ CH_2COOH); 7.54 (d, 2H, J = 7.2 Hz, benzene), 7.25 (m, 3H, benzene). ^{13}C -NMR (400 MHz, $\text{DMSO}-d_6$): 176.64, 175.04, 168.80, 167.97, 154.96, 143.02, 139.20, 135.97, 134.24, 133.40, 132.73, 132.04, 129.58, 126.67, 125.47, 113.68, 109.38, 65.72, 56.02, 53.31, 53.14–52.66, 47.65, 45.31, 45.13, 44.71, 44.55–44.53, 44.40, 44.22, 44.09, 42.74, 33.44, 19.07 (Figures S1–S4).

Results and Discussion

Synthesis and characterisation

The synthesis of MAST is shown in Scheme 1. It was fully characterised by elemental analysis, ^1H -NMR, ^{13}C -NMR, UV-vis and FT-IR spectroscopy, and mass spectrometry (Figures S1–S4). The data for spectral and elemental analysis are in agreement with their chemical structures. MAST is soluble in DMSO, dimethylformamide (DMF), tetrahydrofuran (THF), ethanol (EtOH), methanol (MeOH), acetone and acetonitrile, but is insoluble in water, dichloromethane and tetrachloromethane.

Investigation of the photophysical properties of MAST

It is known that 1,8-naphthalimide derivatives owe their absorption and fluorescence characteristics to the nature of the C-4 substituents on their rings [68,69]. Table 1 shows the fundamental photophysical characteristics [absorption maxima (λ_{A}), fluorescence maxima (λ_{F}), molar extinction coefficient (ϵ_{max}), Stokes shift ($\nu_{\text{A}}-\nu_{\text{F}}$) and quantum fluorescence yield (Φ_{F}) of MAST obtained for seven solvents with varying polarities. The solvents used for assessing the photophysical properties of MAST were DMSO, DMF, THF, EtOH, MeOH, acetone and acetonitrile. These solvents were chosen since they provide excellent solubilities for the chemosensor.

Derivatives of 4-alkylamino-1,8-naphthalimide are strong yellow-greenish-fluorescence-emitting fluorophores with maximal absorption bands of *ca.* 430–440 nm in the blue region, which are attributed to $\pi-\pi^*$ transitions. To the

Table 1 Photophysical properties of MAST (1 μM) in solvents of varying polarities.

Solvent	λ_{F} , nm	λ_{A} , nm	$\epsilon_{\text{max}} \cdot 10^{-4}$, $\text{l mol}^{-1} \text{cm}^{-1}$	$\nu_{\text{A}}-\nu_{\text{F}}$, cm^{-1}	Φ_{F}
DMSO	532	444	3.91	3580.1	0.11
DMF	523	440	7.53	3387.9	0.18
THF	506	429	5.18	3562.5	0.86
EtOH	527	443	2.64	3705.0	0.30
MeOH	528	440	4.65	4317.1	0.40
Acetone	516	429	4.09	4429.7	0.49
Acetonitrile	517	430	3.75	3752.3	0.64

naked eye, MAST displayed a yellow colour in each of these solvents; it emitted a strong greenish fluorescence under UV light. The fluorescence and absorption bands obtained were 506–532 and 429–444 nm, respectively.

Increasing the solvent polarity usually leads to a bathochromic shift of the maximum wavelength band. As was speculated, DMSO, since it has the highest polarity, displayed the highest absorption and fluorescence wavelengths (λ_A and λ_F) and THF, with the lowest polarity, yielded the lowest values of λ_A and λ_F . The range in the values of the molar extinction coefficient obtained was $11.264\text{--}7.53 \times 10^{-4} \text{ dm}^3 \text{ mol}^{-1} \text{ cm}^{-1}$, which is representative of typical 1,8-naphthalimide derivatives.

Stokes shift is a key photophysical parameter of fluorescent compounds that conveys the variations in the properties and structure of the fluorophores between the ground state, S_0 , and the first excited state, S_1 [70]. Based on Eqn 1 [71], the values of the Stokes shift obtained were sufficiently high over a range of $3387.9\text{--}4429.7 \text{ cm}^{-1}$, which is in agreement with the data obtained for other blue-emitting 1,8-naphthalimide derivatives [72]:

$$(\nu_A - \nu_F) = \left(\frac{\lambda_F - \lambda_A}{\lambda_A \lambda_F} \right) \times 10^7 \quad (1)$$

This attests that Stokes shift values are usually high in polar solvents, where there are favourable formations of hydrogen bond and dipole interactions [73].

The ability of molecules to emit absorbed light is portrayed quantitatively via Φ_F , the ratio of the number of photons emitted to the number of photons absorbed; Φ_F was obtained for each of the solvents using Eqn 2 [74] with N-butyl-4-n-butylamino-naphthalimide ($\Phi_F = 0.81$ in EtOH) deployed as the standard:

$$\phi_F = \phi_r \frac{S_s A_r n_s^2}{S_r A_s n_r^2} \quad (2)$$

where Φ_r denotes Φ_F of the standard, A_r and A_s are the absorbances of the standard and the sample at their respective excited wavelengths, S_r and S_s are the respective integrated band areas of the standard and the sample, and n_r and n_s represent the respective solvent refractive indexes of the standard and the sample.

The quantum yield decreases as solvent polarity increases; THF has the highest and DMSO has the lowest value. This could be due to quenching of the PET, a phenomenon often observed in non-polar media. This, in turn, led to the restoration of the fluorescence emission [75].

Investigation of the effect of pH on the sensing properties of MAST

The next step was to gain an understanding of the influence of operating pH on the sensing properties of MAST, to determine the pH at which the proposed chemosensor performs best. It is known that the presence or absence of protons in the vicinity of fluorescent chemosensors influences their sensing behaviour, which affects their sensitivities in detecting an analyte.

The fluorescence spectra of MAST was monitored as a function of variation in pH by conducting pH titration of MAST with solutions of different pH values; pH was modulated stepwise with 0.1 M hydrogen chloride and/or 0.1 M sodium hydroxide over a pH range of 0.54–13.58. As shown in Figure 1a, upon acidification large fluorescence enhancements were observed over the pH range of 0.54–1.53, indicating that the fluorescence emission is highly dependent on pH in the acidic region. This is attributed to the protonation of the secondary amine nitrogen on the naphthalimide ring [3], which in turn weakens its interaction with Cu(II), leading to an enhanced turned-on fluorescence intensity. Interestingly, only minor changes (<2.46% variation) in the fluorescence intensity of the chemosensor occurred over the pH range of 1.53–8.62; the determination of Cu(II) ions with MAST was not significantly affected by variations in pH in this region. This therefore proves that the synthesised chemosensor could facilitate the monitoring of Cu(II) ions over a wide pH range, and that it can function as a chemosensor in both low and medium pH-value media. From the results, a pH of 7.4 was chosen as the optimal pH at which the proposed chemosensor works best (Figure 1b).

Conversely, the fluorescence intensity of the chemosensor suffered a marked decrease (turned-off fluorescence) over the pH range 9.58–13.58. This is attributed to the deprotonation of the hydroxyl functional group in the

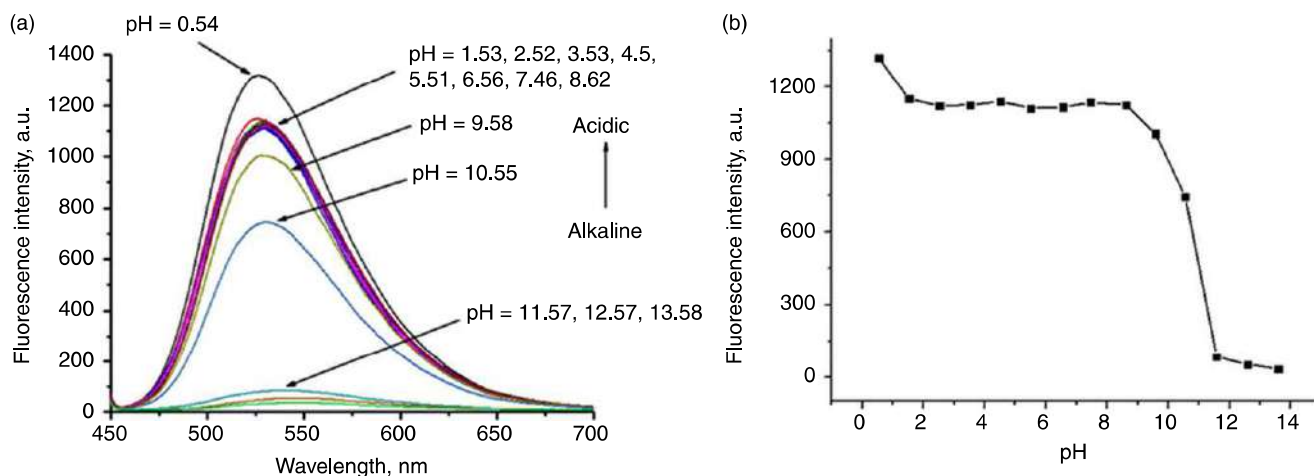


Figure 1 (a) Influence of pH on the fluorescence spectra of chemosensor MAST (0.5 μM) in DMSO solution. The pH was modulated by adding 0.1 M HCl and/or 0.1 M NaOH in double-distilled water. (b) The changes at 529 nm as a function of pH [Colour figure can be viewed at wileyonlinelibrary.com]

receptor molecule, which in turn leads to an extension of the conjugation, thereby reducing the fluorescence intensity.

UV-vis and fluorescence titration of MAST towards Cu(II) ions

To investigate the binding affinity of MAST with Cu(II) ions, UV-vis and fluorescence titration experiments were conducted. The spectra behaviour of MAST (1 μM) was recorded in DMSO/HEPES buffer solution (v/v, 1:1; pH = 7.4). The free chemosensor displays an absorption band centered at $\lambda_{\text{max}} = 438 \text{ nm}$ ($\epsilon_{\text{max}} = 1.41 \times 10^4 \text{ dm}^3 \cdot \text{mol}^{-1} \cdot \text{cm}^{-1}$), which is responsible for its sharp yellow colour in solution. The chemosensor also emits an intensive greenish fluorescence at $\lambda_{\text{F}} = 529 \text{ nm}$ ($\Phi_{\text{F}} = 0.107$), a representative feature of Φ_{F} of certain fluorescent and 4-amino-1,8-naphthalimide compounds [76–79]. Details of the titration studies conducted follow.

Figure 2a shows the UV-vis absorption titration profile of MAST with various concentrations of Cu(II) solution. As the concentration of Cu(II) ions was increased to 1.8 μM , the

intensity of the main absorption peak at 438 nm underwent a gradual decrease, along with dose-dependent absorbance quenching of the chemosensor towards Cu(II) ions (Figure 2b).

Utilising procedures similar to those of UV-vis titration, further insights into the sensing properties of MAST were obtained by investigating the fluorescence titration reaction curve of the changes in the fluorescence spectra of the chemosensor towards different amounts of Cu(II) ions (Figure 3a). The fluorescence titration of the reaction of MAST with Cu(II) indicates that the fluorescence intensity of the chemosensor was quenched in a steady manner in response to the gradual increase in the concentration of the Cu(II) added to the DMSO/HEPES buffer solution (v/v, 1:1; pH = 7.4), with a bathochromism of *ca.* 3 nm observed. This can be explained by the complexation of the Cu(II) ions, which reduces the electron-releasing ability of the nitrogen atom in the receptor molecule that reverses the PET process, and the intrinsic paramagnetic property of Cu(II) ions. The fluorescence quenching indicates that the chemosensor can act effectively as a turn-off chemosensor

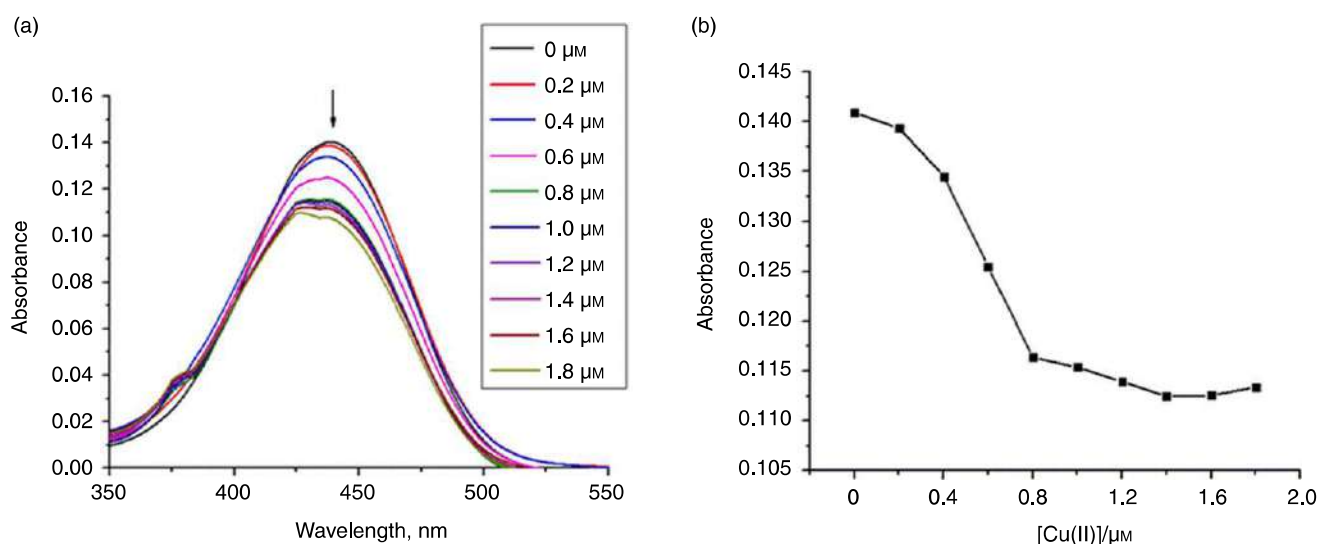


Figure 2 (a) Changes in the absorption spectra of MAST (1 μM) in DMSO/HEPES buffer solution (v/v, 1:1; pH 7.4) in the presence of different amounts of Cu²⁺ (0–1.8 μM). (b) Absorbance at 438 nm as a function of Cu²⁺ concentrations [Colour figure can be viewed at wileyonlinelibrary.com]

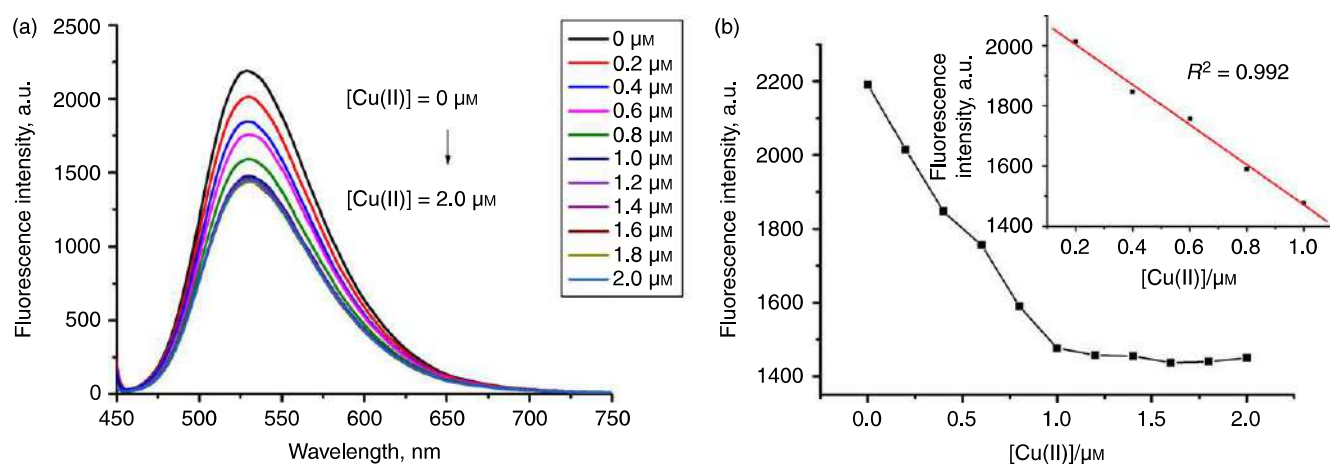


Figure 3 (a) Fluorescent spectra of chemosensor MAST (1 μM) in DMSO/HEPES buffer solution (v/v, 1:1; pH 7.4) in the presence of different amounts of Cu²⁺ (0–2.0 μM). (b) Fluorescence intensity at 529 nm as a function of Cu²⁺ concentrations. Excitation was performed at 435 nm; Insert figure: linear fitting of the data [Colour figure can be viewed at wileyonlinelibrary.com]

towards Cu(II) ions. As depicted in Figure 3b, the dose-dependent fluorescence quenching of the chemosensor towards Cu(II) ions showed good linearity up to $1.0\ \mu\text{M}$ of the Cu(II) solution, enabling a range of $0.2\text{--}1\ \mu\text{M}$ to be set as the threshold for the determination of Cu(II) ions. This encouraging outcome establishes the effectiveness of the proposed chemosensor to detect quantitatively relevant concentrations of Cu(II) ions.

Spectroscopic assessment of MAST in the presence of various metal ions

The fluorescence response of MAST to various metal ions, namely, Na(I), K(I), Ca(II), Mg(II), Al(III), Pb(II), Fe(III), Ni(II), Zn(II), Hg(I), Ag(I), Co(II), Mn(II), Cd(II) and Cu(II) in DMSO/HEPES buffer solution (v/v, 1:1; pH = 7.4) under excitation at 435 nm, was investigated. In the absence of the metal ions, the chemosensor ($0.5\ \mu\text{M}$) showed substantial fluorescence strength (an intensive greenish fluorescence) at 529 nm. As can be seen in Figure 4, when $5\ \mu\text{M}$ of each of the metal ions was introduced into the DMSO/HEPES buffer solution (v/v, 1:1; pH = 7.4), only the Cu(II) ions were able to quench the fluorescence of the chemosensor. This observation occurred instantly affording a fluorescence decrease of 83.67% or 6.12-fold quenching of fluorescence ($\Phi_F = 0.018$). This is due to the coordination of the paramagnetic Cu(II) to the receptor unit of the chemosensor, which affected the fluorescence turn-off behaviour [63–66]. These results indicate that MAST could be used as a Cu(II) selective fluorescent sensor.

Selectivity study of MAST for Cu(II) ions over competitive metal ions

Whether a proposed compound will act as an ideal chemosensor or not is largely a matter of its selectivity. The selectivity of the proposed chemosensor was studied via fluorescence competitive experiments to examine the effect of other metal ions on its sensing properties: $0.5\ \mu\text{M}$ of Cu(II) ions were mixed with $0.5\ \mu\text{M}$ of each of the various interfering metal ions, namely, Na(I), K(I), Ca(II), Mg(II), Al(III), Pb(II), Fe(III), Ni(II), Zn(II), Hg(I), Ag(I), Co(II), Mn(II) and Cd(II). As shown in Figure 5, the presence of other cations did not have any significant fluorescence signal interference effect on MAST–Cu(II), and the unique

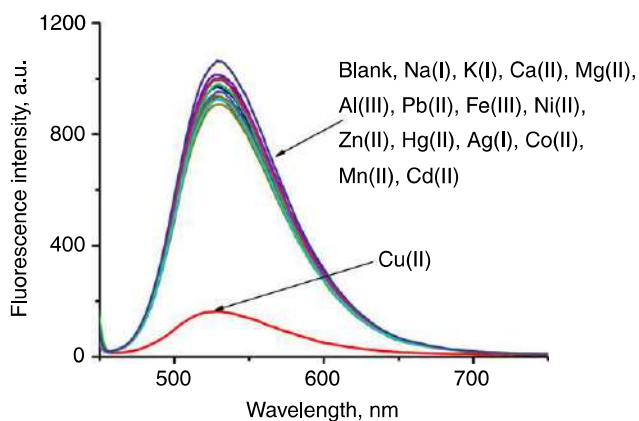


Figure 4 Fluorescence spectra of chemosensor MAST ($0.5\ \mu\text{M}$) in DMSO/HEPES buffer solution (v/v, 1:1; pH 7.4) in the presence of different competitive metal ions ($5\ \mu\text{M}$) [Colour figure can be viewed at [wileyonlinelibrary.com](#)]

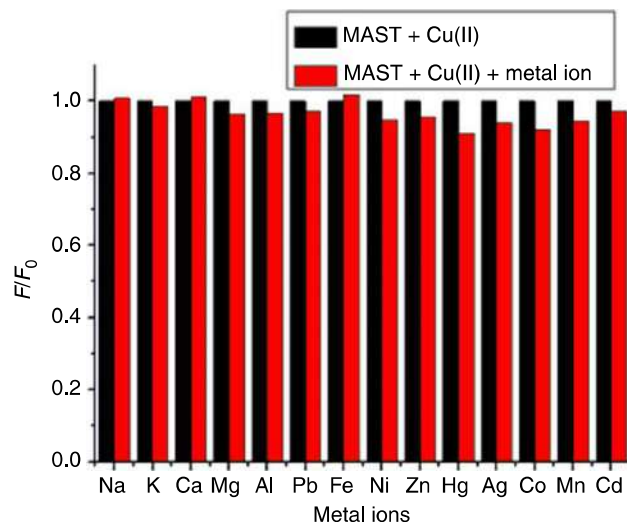


Figure 5 Fluorescence response of chemosensor MAST ($0.5\ \mu\text{M}$) for Cu^{2+} ($0.5\ \mu\text{M}$) in the presence of other metal ions ($0.5\ \mu\text{M}$) in DMSO/HEPES buffer solution (v/v, 1:1; pH 7.4) [Colour figure can be viewed at [wileyonlinelibrary.com](#)]

spectral and colour changes were retained. Results indicate that MAST could be used as a selective fluorescent sensor for detecting Cu(II) ions in the presence of other competing metal ions, and that the detection of Cu(II) ions by MAST was unaffected by coexistent metal cations.

Determination of the stoichiometric ratio and binding constant

The binding stoichiometry of MAST towards Cu(II) ions in the MAST–Cu(II) complex was estimated by continuous variation using Job plot analysis [80]. As can be seen from Figure 6, the peak of the Job plot is equivalent to the mole fraction at 0.5, indicating a 1:1 stoichiometric interaction between Cu(II) and MAST.

Furthermore, the association constant (K_a) of MAST, a critical parameter in determining the performance efficiency of a synthesised compound as a chemosensor, was calculated by 1:1 stoichiometry using the Benesi-Hildebrand equation [81]:

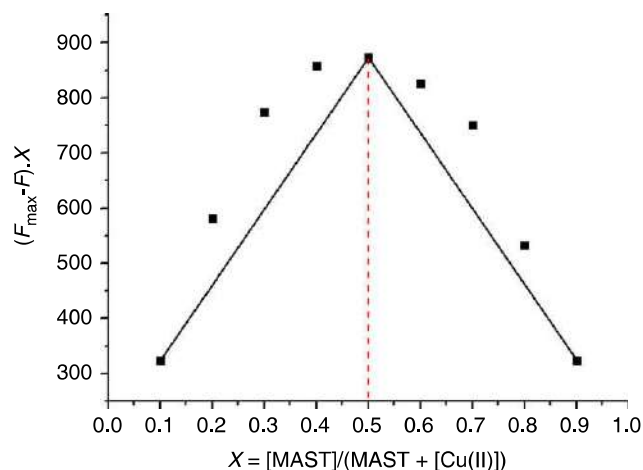


Figure 6 Job plot for the determination of the binding ratio of chemosensor MAST and Cu(II) in DMSO/HEPES buffer solution (v/v, 1:1; pH 7.4) [Colour figure can be viewed at [wileyonlinelibrary.com](#)]

$$\frac{1}{F - F_0} = \frac{1}{K_a(F_{\min} - F_0)[\text{Cu}^{2+}]^n} + \frac{1}{F_{\min} - F_0} \quad (3)$$

where F is the fluorescence intensity at 529 nm for any given concentration of Cu(II), F_0 is the fluorescence intensity at 529 nm in the absence of Cu(II), $F_{\min}$ is the minimum fluorescence intensity at 529 nm in the presence of Cu(II) in solution, and n is the stoichiometric mole ratio (1 in this case). K_a was evaluated by plotting $1/(F - F_0)$ vs $1/[\text{Cu(II)}]$ (Figure 7) and was calculated to be $1.37 \times 10^6 \text{ M}^{-1}$. This correlates with those values obtained for other Cu(II) chemosensors [82–85]. Data were linearly fitted according to Eqn 3 and K_a was derived from the slope of the line. The large K_a of MAST for Cu(II) ions is further confirmation of the existence of strong coordination in the MAST–Cu(II) complex.

Determination of the detection limit of MAST

The fluorescence quenching for the turn-off phenomenon was evaluated quantitatively by observing the Stern-Volmer plot of the ratio of the fluorescence intensities vs the concentration of Cu(II) ions (Figure 8) based on [86]:

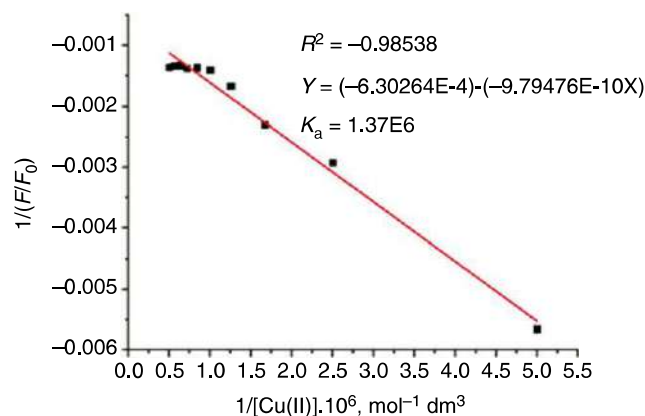


Figure 7 Benesi-Hildebrand plot for the determination of the association constant of complex [Cu(MAST)] based on a 1:1 stoichiometry for the association between MAST and Cu(II) ion [Colour figure can be viewed at [wileyonlinelibrary.com](#)]

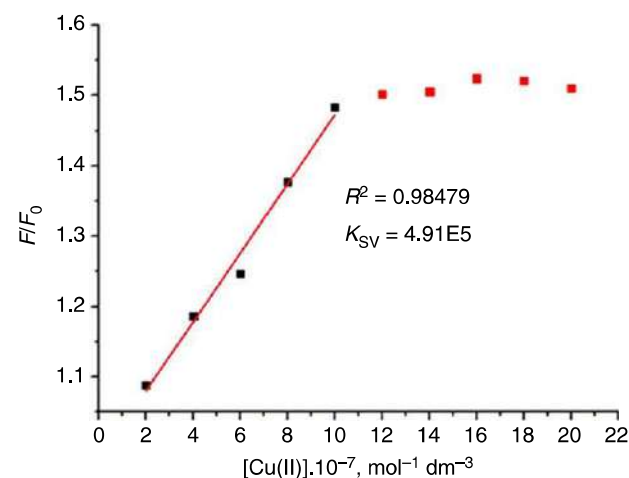


Figure 8 Stern-Volmer plot of the ratio of the fluorescence intensities F_0/F (where F_0 is the fluorescence intensity of the blank and is equal to 2191) of chemosensor MAST against the increasing concentration of Cu^{2+} [Cu^{2+}] = 0–2.0 μM] [Colour figure can be viewed at [wileyonlinelibrary.com](#)]

$$\frac{F_0}{F} = 1 + K_{\text{sv}}[\text{Q}] \quad (4)$$

where F_0 is the initial fluorescence intensity, F is the fluorescence intensity measured in the presence of the analyte, $[\text{Q}]$ is the concentration of the quencher [Cu(II) in this case] and K_{sv} is the Stern-Volmer constant. When F_0/F was plotted vs $[\text{Q}]$, the plot was a straight line, with K_{sv} being equivalent to the value of the slope.

The detection limit (DL), a distinctive feature any proposed compound must possess to qualify as a valid chemosensor, was also obtained [87]. The DL of MAST towards Cu(II) was calculated using:

$$\text{DL} = 3\sigma/K_{\text{sv}} \quad (5)$$

where σ is the standard deviation of the blank solution and K_{sv} is the slope derived from the calibration plot. DL was calculated to be $1.69 \times 10^{-8} \text{ M}$, which is significantly lower than the respective WHO and EPA regulated limits of $31.5 \times 10^{-6} \text{ M}$ and $20 \times 10^{-6} \text{ M}$ for Cu(II) ions in drinking water [34,43]. This result confirms that the proposed chemosensor MAST has the potential to monitor Cu(II) ions in acceptably low concentrations in environmental systems.

All the described experiments yielded robust results which fit with the synthesis of a Cu(II)-selective fluorescence turn-off chemosensor. Therefore, the proposed chemosensor MAST fulfils the criteria for highly selective Cu(II) sensing in competitive media of various physiologically and environmentally relevant metal ions.

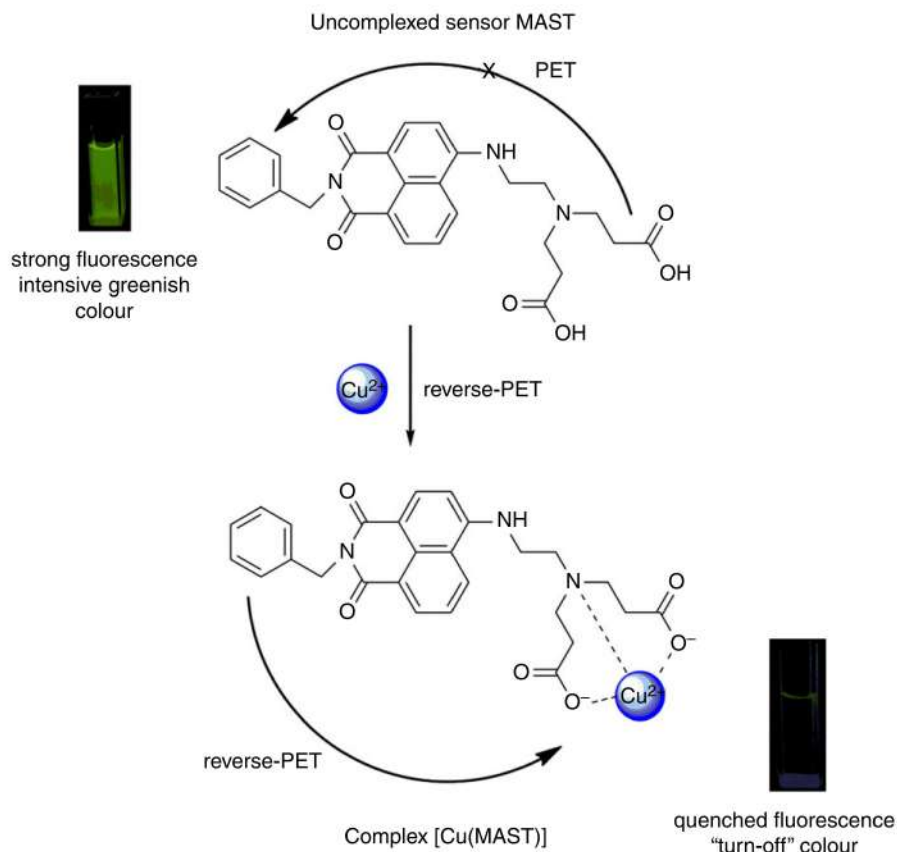
Proposed sensing mechanism

The results of Job's plot, fluorescence titration and absorbance titration indicate that MAST may form a chelate with Cu(II) ions through interactions with ethanediamine nitrogen and oxygen of the carboxyl group, forming a 1:1 complex (Scheme 2). This binding mode was further established by mass spectrometry. When a solution of Cu (Ac) $_2 \cdot \text{H}_2\text{O}$ was added to an equivalent solution of MAST, two main peaks appeared (Figure S5). The first peak at $m/z = 490.1853$ corresponded to $[\text{MAST} + \text{H}]^+$, and the second peak at $m/z = 552.1712$ corresponded to $[\text{MAST} - \text{H} + \text{Cu}^{2+}]^+$, thereby confirming the evidence of a 1:1 binding mode.

Scheme 2 shows the detailed illustration of the quenching mechanism. The medium fluorescence of MAST is because the iminodipropionic group did not have a lone pair of sufficient energy to cause PET in the molecule. Cu(II) was paramagnetic with an unfilled d shell and thereby could quench the emission of the fluorophore near it via electron transfer. The quenching response of MAST towards Cu(II) ions could be ascribed to the reverse-PET mechanism, that is, the electron transfer from 1,8-naphthalimide moiety to an electron-deficient carboxyl group occurring by metal ion complexation. [29,63–66].

Response time

In addition to high sensitivity and selectivity, a short response time is necessary for a fluorescent sensor to dynamically monitor Cu(II) in environmental samples in real time. Figure 9 shows the time duration for the fluorescence intensity of the chemosensor MAST (0.5 μM , DMSO/



Scheme 2 Probable binding mode of fluorescence quenching of chemosensor MAST with Cu²⁺. Inset: the concomitant “turn off” color change under UV light [Colour figure can be viewed at wileyonlinelibrary.com]

HEPES buffer, pH = 7.4) treated with 0.5 μM of Cu(II) at room temperature. This indicates that the reaction of MAST with Cu(II) can be carried out under these conditions within 3 min. Also, once a plateau is reached, the fluorescence intensity at 529 nm remains unchanged for the rest of the experiment, indicating that MAST is stable to the medium, light and air.

Real-world application of MAST for Cu(II) quantification

Chemosensors that facilitate the practical monitoring of analytes are extremely useful. The effectiveness of the synthesised chemosensor MAST was evaluated by applying

it to detect Cu(II) in actual water samples. The water samples were sourced from the Yellow River (indigenous to Lanzhou City in North West China) and tap water from Lanzhou Jiaotong University was used. The environmental water samples were filtrated and showed that no Cu(II) ions were present in them. The solutions of Cu(II) at concentration levels of 2×10^{-7} M, 3×10^{-7} M and 4×10^{-7} M were spiked in all of the real samples. The pH values of these samples were adjusted with the DMSO/HEPES buffer solution (v/v, 1:1; pH = 7.4) and analysed with MAST. The recoveries of the spiked Cu(II) samples were calculated and summarised (Table 2). Analysis of the results gave recoveries for a range of 98–105.8%, indicating that MAST could facilitate the detection of Cu(II) ions in real water samples.

Conclusions

A new, highly potent 1,8-naphthalimide-based fluorescence turn-off chemosensor, MAST, was successfully synthesised,

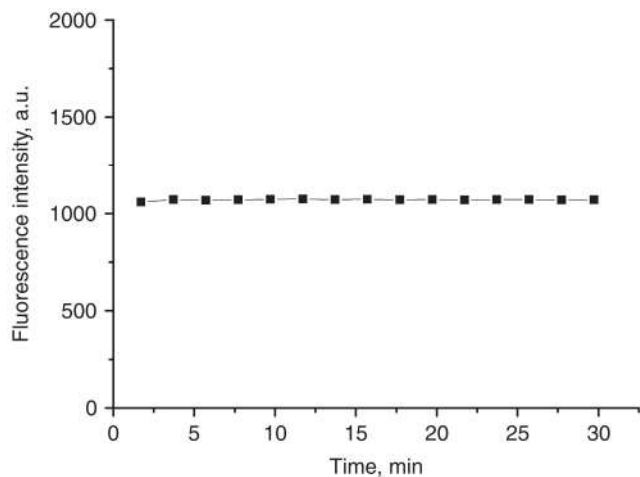


Figure 9 Time course for the fluorescence response of chemosensor MAST (0.5 μM , DMS/HEPES buffer, pH 7.4) to Cu²⁺ (0.5 μM) at 25 °C

Table 2 Results of the analysis of Cu(II) ions in different water samples using MAST

Water sample	Cu(II) spiked, 10^{-7} M	Cu(II) recovery, mean SD, 10^{-7} M	Recovery, %
Yellow river	2	2.02 ± 0.13	101.0
	3	3.14 ± 0.20	104.7
	4	3.98 ± 0.16	99.5
Tap	2	1.96 ± 0.03	98.0
	3	3.07 ± 0.25	102.3
	4	4.23 ± 0.32	105.8

and its sensing properties towards Cu(II) ions were evaluated. Via a dampened fluorescence mechanism, MAST had a monopoly on sensing Cu(II) ions in a DMSO/HEPES buffer solution (v/v, 1:1; pH = 7.4) in the presence of other competitive metal ions, namely, Na(I), K(I), Ca(II), Mg(II), Al(III), Pb(II), Fe(III), Ni(II), Zn(II), Hg(I), Ag(I), Co(II), Mn(II) and Cd(II). DL of the chemosensor for the monitoring of Cu(II) ions reached 1.69×10^{-8} M, making it highly usable for the quantitative analysis of Cu(II) ions. The results obtained from the application of the sensor to track Cu(II) ions in environmental water samples prove its potency as an ideal chemosensor.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Figure S1 ¹H-NMR spectrum of MAST

Figure S2 ¹³C-NMR spectrum of MAST

Figure S3 HRMS spectrum of MAST

Figure S4 FT-IR spectrum of MAST

Figure S5 HRMS spectrum of [Cu(Ac)₂·H₂O + MAST]